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B																									
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D																									
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F																									
UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-20 & MVD-21 20kV Boiler Plant SCHEMATIC DIAGRAM DRC BUS PT H04																									
				Date Prep.		15.11.2018		PROJECT TITLE				ABB		EPMV		COVER SHEET		Scale				= H04			
				Prepared		YR		MV Switchgear for MNA Serang Plant				PT. ABB Sakti Industri		MVD-20 & MVD-21 20kV Boiler Plant SCHEMATIC DIAGRAM DRC BUS PT		Customer		PT Multimas Nabati Asahan		& 308					
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F					Date Prep.	15.11.2018	PROJECT TITLE		ABB EPMV	Index of Sheet		Scale		= H04	F	
					Prepared	YR	MV Switchgear for MNA Serang Plant			MVD-20 & MVD-21 20kV Boiler Plant		Customer	PT Multimas Nabati Asahan	& 308		
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A	-BGI31		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION										-CA	CAPACITORS										A
	-BGI32		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION										-CA1	CAPACITOR FOR FRONT VENTILATOR										
	-BGI33		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)										-CA2	CAPACITOR FOR REAR VENTILATOR										
	-BGI41...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION										-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.										
	-BGI42		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION										-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT										
	-BGI43...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION										-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE										
	-BGI51		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION										-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT										
	-BGI52		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION										-EB1	HEATER LOCATED IN CABLE COMPARTMENT										
	-BGI53		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION										-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT										
	-BGK		POSITION SWITCH OF KEY LOCK										-EB3	HEATER LOCATED IN MOTOR CABINET										
B	-BGL1		POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1										-EB4	HEATER FOR VENTILATION TEST										B
	-BGS1...-BGS4		POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS										-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.										
	-BGS5		PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION										-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE										
	-BGT1		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION										-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE										
	-BGT2		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION										-FA1	SURGE ARRESTER LOCATED ON PHASE L1										
	-BGT3		POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION										-FA2	SURGE ARRESTER LOCATED ON PHASE L2										
	-BGT4		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION										-FA3	SURGE ARRESTER LOCATED ON PHASE L3										
	-BGT5		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION										-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS										
	-BGT6		POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION										-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1										
	-BGT7		POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION										-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2										
C	-BGT8		POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION										-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3										C
	-BGT11		PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION										-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER										
	-BGT12		PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION										-FCM2....-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS										
	-BGV1		POSITION SWITCH OF FRONT VENTILATOR										-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1										
	-BGV2		POSITION SWITCH OF REAR VENTILATOR										-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2										
	-BM		HYGROSTAT										-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3										
	-BPA4		PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING										-FCT	THERMAL OVERLOAD RELEASES										
	-BPA5		PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING										-GA	GENERATORS										
	-BPA6		PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING										-GQ1	FRONT VENTILATOR										
	-BPA7		PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING										-GQ2	REAR VENTILATOR										
D	-BPS1		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER										-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK										D
	-BPS2		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2										-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE										
	-BPS3		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3										-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE										
	-BR		FLAME DETECTORS, SMOKE DETECTORS										-KFA3...-KFA9	AUXILIARY RELAYS OR CONTACTORS										
	-BT		THERMOSTAT										-KFC	CLOSING RELAYS OR CONTACTORS										
	-BT1		THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS										-KFI	INTEGRATED CIRCUITS										
	-BT2		THERMOSTAT FOR ENERGIZATION OF VENTILATORS										-KFL	LOCKOUT RELAY										
	-BT3		THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM										-KFM	MICROPROCESSORS										
	-BU		BUCHHOLZ RELAY										-KFN	ANTIPUMPING RELAYS										
	-BX1		UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING										-KFO	OPENING RELAYS OR CONTACTORS										
E	-BX2		ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING										-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)										E
	-BZ,-BZ4		MULTIFUNCTION SENSORS										-KFR	RECLOSING RELAYS										
	-BZ1		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1										-KFS	SYNCHRONIZING DEVICES										
	-BZ2		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2										-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS										
	-BZ3		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3										-KFZ	CONTROLLERS										
													-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER										
													-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD										
													-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1										
													-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2										
													-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS										
F					Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB	EPMV	REFERENCE DESIGNATIONS MVD-20 & MVD-21 20kV Boiler Plant		Scale			= H04	F							
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A	-MAE		MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE		-QBH	MANUAL CIRCUIT-BREAKERS		A	
	-MAE1		MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2			-QBL			LINKS
	-MAE8		MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS		-QBM		MINIATURE SWITCH-DISCONNECTORS		
					-QBS		SWITCH-DISCONNECTORS		
B	-MAS		MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING		-QBT		TRUCK		B
	-MAT		MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT		-QCA		EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING		
	-MBC		CLOSING RELEASE OF CIRCUIT-BREAKER		-QCE		EARTHING SWITCH		
	-MBC4		CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS		-QCE1		EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)		
	-MBO1		FIRST OPENING RELEASE OF CIRCUIT-BREAKER		-QCE2		EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)		
	-MBO2		SECOND OPENING RELEASE OF CIRCUIT-BREAKER		-QQ		CLUTCHES		
	-MBO3		OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER		-RAA		ANTIFERRORESONANCE RESISTOR		
					-RAD		DIODES		
C	-MBO4		OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS		-RAI		INDUCTORS		C
	-MBU		UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER		-RAR		RESISTORS		
	-PFB		BLUE SIGNAL LAMPS		-RF		FILTERS		
	-PFF		FLAG RELAYS		-RLE1		ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING		
	-PFG		GREEN SIGNAL LAMPS		-RLE2		ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT		
	-PFR		RED SIGNAL LAMPS		-RLE3		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE		
	-PFS		SHORT CIRCUIT INDICATORS		-RLE4		ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION		
	-PFV		VOLTAGE INDICATORS		-RLE5		ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD		
	-PFV1		VOLTAGE INDICATORS ON FEEDER SIDE		-RLE6		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1		
	-PFV2		VOLTAGE INDICATORS ON BUSBAR SIDE		-RLE7		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2		
D	-PFW		WHITE SIGNAL LAMPS		-RLE8		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2		D
	-PFX		CROSS INDICATORS, ELECTROMECHANICAL INDICATORS		-RLE9		ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS		
	-PFY		YELLOW SIGNAL LAMPS		-RLE11		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4		
	-PGA		AMMETERS		-RLE12		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5		
	-PGC		COUNTERS		-SFA		AMMETRIC SWITCHES		
	-PGF		FREQUENCYMETERS		-SFC		CONTROL SWITCHES, CLOSING PUSH-BUTTONS		
	-PGH		HOURMETERS		-SFL		LOCKING CONTACTS		
	-PGI		PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE		-SFO		OPENING PUSH-BUTTONS		
	-PGI1		HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)		-SFR		RESET PUSH-BUTTONS		
					-SFS		SELECTOR SWITCHES		
E	-PGI2		HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)		-SFT		TEST PUSH-BUTTONS		E
	-PGJ		ACTIVE ENERGY METERS		-SFU		UNLOCKING PUSH-BUTTONS		
	-PGK		REACTIVE ENERGY METERS		-SFV		VOLTMETRIC SWITCHES		
	-PGM		MULTIFUNCTION INDICATORS		-TA		POWER TRANSFORMERS		
	-PGP		POWER-FACTOR METERS		-TB		CONVERTER		
	-PGQ		VARMETERS		-TB1		RECTIFIER FOR FIRST OPENING RELEASE		
	-PGS		SYNCHRONOSCOPES		-TB2		RECTIFIER FOR SECOND OPENING RELEASE		
	-PGV		VOLTMETERS		-TB3		RECTIFIER FOR CLOSING RELEASE		
	-PGW		WATTMETERS		-TB4		RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING		
	-PJ		ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)		-TB5		RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT		
	-QAB		CIRCUIT-BREAKERS		-TB6		RECTIFIER FOR UNDERVOLTAGE RELEASE		
	-QAC		CONTACTORS (FOR POWER)		-TB7		ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER		
	-QBD		DISCONNECTORS						
	-QBD1		FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						
	-QBD2		REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						
	-QBD3		FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						
	-QBD4		TWO-WAY DISCONNECTOR						
	-QBD5		TWO-WAY DISCONNECTOR ON BUSBAR RISER						

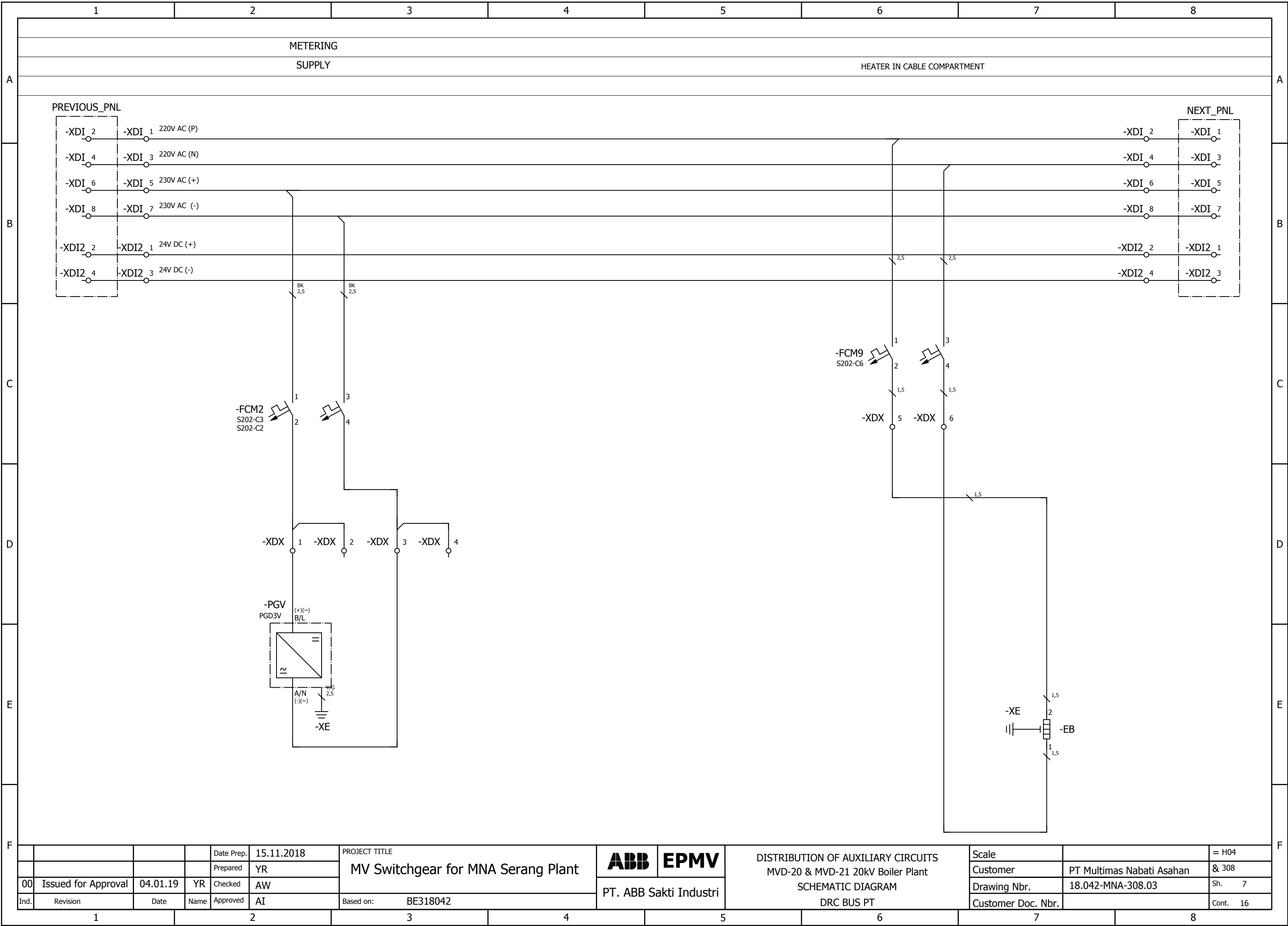
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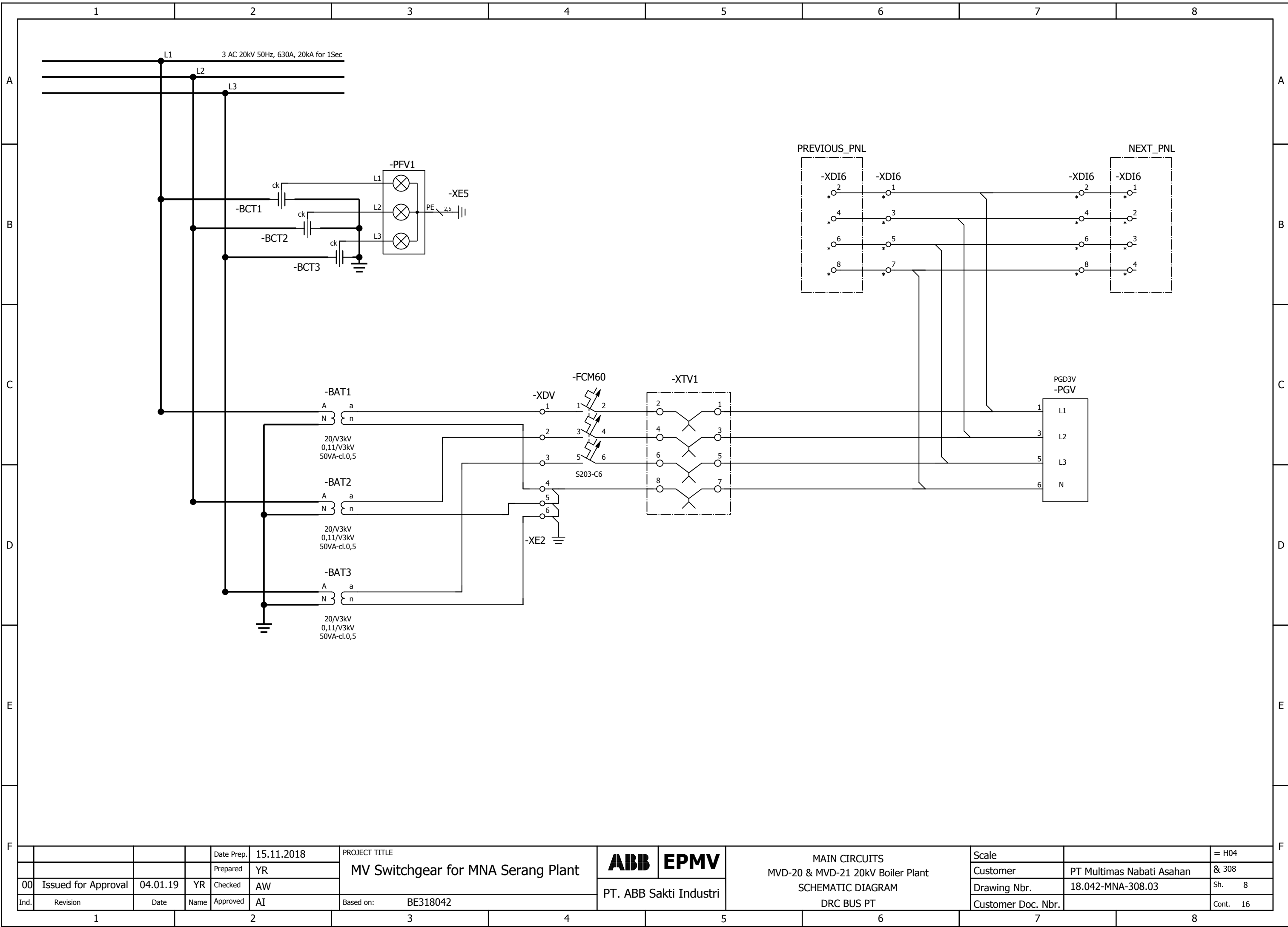
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A	-TFA				ACTIVE POWER TRANSDUCERS				-XD11				TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS				A
	-TFC				CURRENT TRANSDUCERS				-XD12				TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING				
	-TFF				FREQUENCY TRANSDUCERS				-XD13				TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION				
	-TFJ				ACTIVE ENERGY TRANSDUCERS				-XD14				TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS				
	-TFK				REACTIVE ENERGY TRANSDUCERS				-XD16				TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS				
	-TFM				MULTIFUNCTION TRANSDUCERS												
	-TFP				POWER-FACTOR TRANSDUCERS				-XD17				TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS				
	-TFQ				REACTIVE POWER TRANSDUCERS				-XD18				TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS				
B	-TFS				SIGNAL CONVERTERS				-XD19				TERMINAL BLOCK FOR SPECIAL USE				B
	-TFV				VOLTAGE TRANSDUCERS				-XDJ				TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD				
	-WA				BUSBARS				-XDJ1				TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS				
	-WBC				POWER CABLES				-XDM				SEALABLE TERMINAL BLOCK FOR MEASUREMENT				
	-WE				EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.				-XDS				CURRENT SOCKETS				
	-WGC				CONTROL CABLES				-XDT				TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB				
	-WGS				COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)				-XDU				CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK				
	-WH				OPTICAL FIBRES				-XDU1				TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK				
	-XDA				TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS				-XDV				TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS				
	-XDA1				CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1				-XDV1				CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1				
	-XDA2				CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2				-XDV2				CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2				
	-XDA3				CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3				-XDV3				CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3				
C	-XDA5				CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER				-XDV4				CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR				C
	-XDB				CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS				-XDX				SUPPORT TERMINAL BLOCK				
	-XDB1				CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)				-XE1				EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE				
	-XDB2				CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER				-XE2				EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE				
	-XDB10...-XDB89				CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS				-XE5				EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE				
	-XDB7				CONNECTOR FOR TRUCK POSITION CONTACTS				-XE6				EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE				
	-XDB91				CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR				-XE7				EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE				
	-XDB92				CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR				-XE8				EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE				
	-XDB93				CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR				-XE9				EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT				
	-XDB95				CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET												
D	-XDC				CUSTOMER TERMINAL BLOCK												D
	-XDC1				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER												
	-XDC2				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR												
	-XDC3				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT												
	-XDC4				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS												
	-XDC5				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER												
	-XDE				TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE												
	-XDE1				TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE												
	-XDE2				TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2												
	-XDE3				TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2												
	-XDF				TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS												
	-XDF1				CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR												
	-XDF2				CONNECTOR FOR REAR VENTILATOR CIRCUITS												
	-XDF3				CONNECTOR FOR FRONT VENTILATOR CIRCUITS												
F	-XDH				TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)												F
	-XDI				TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)												

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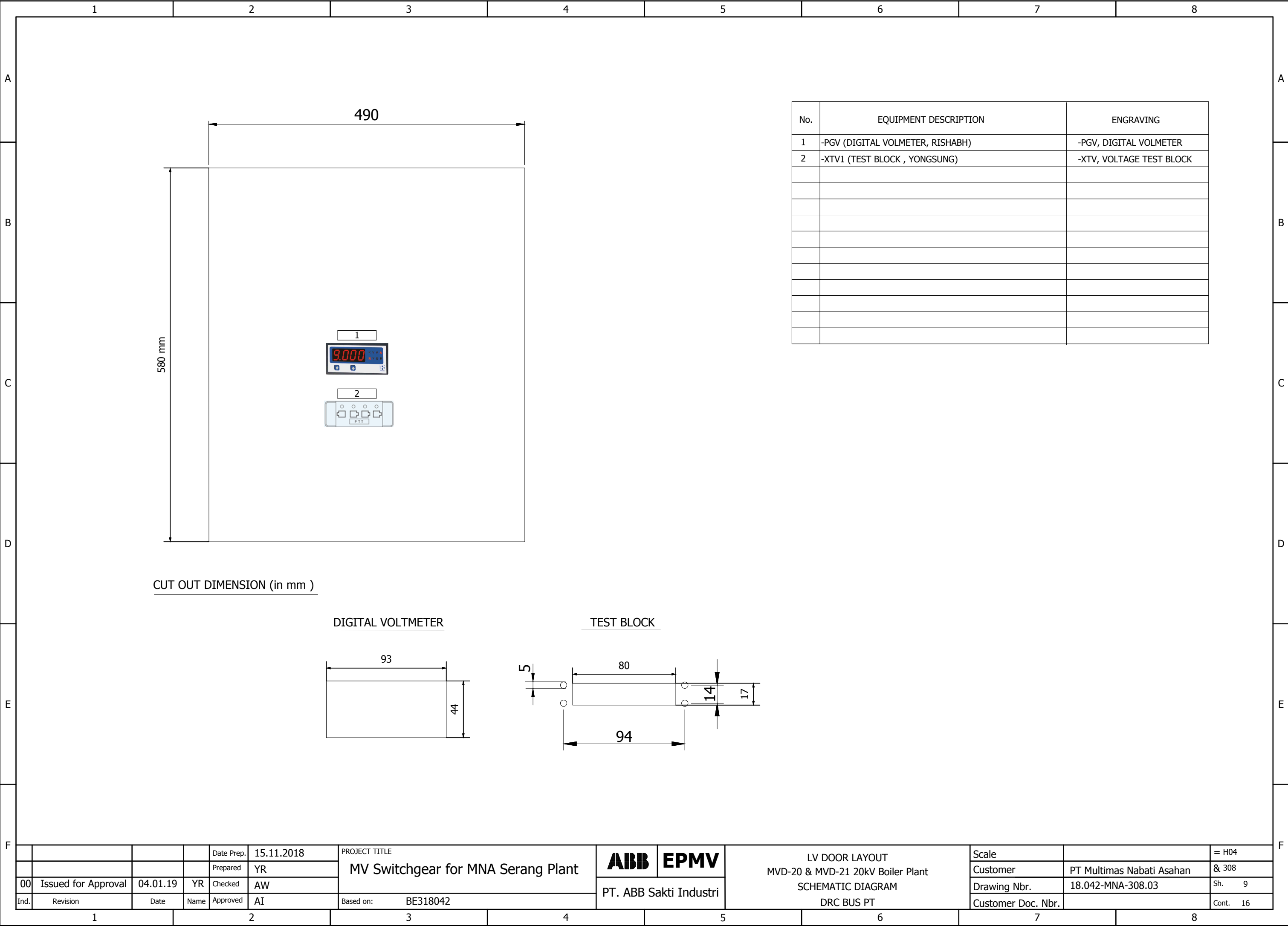


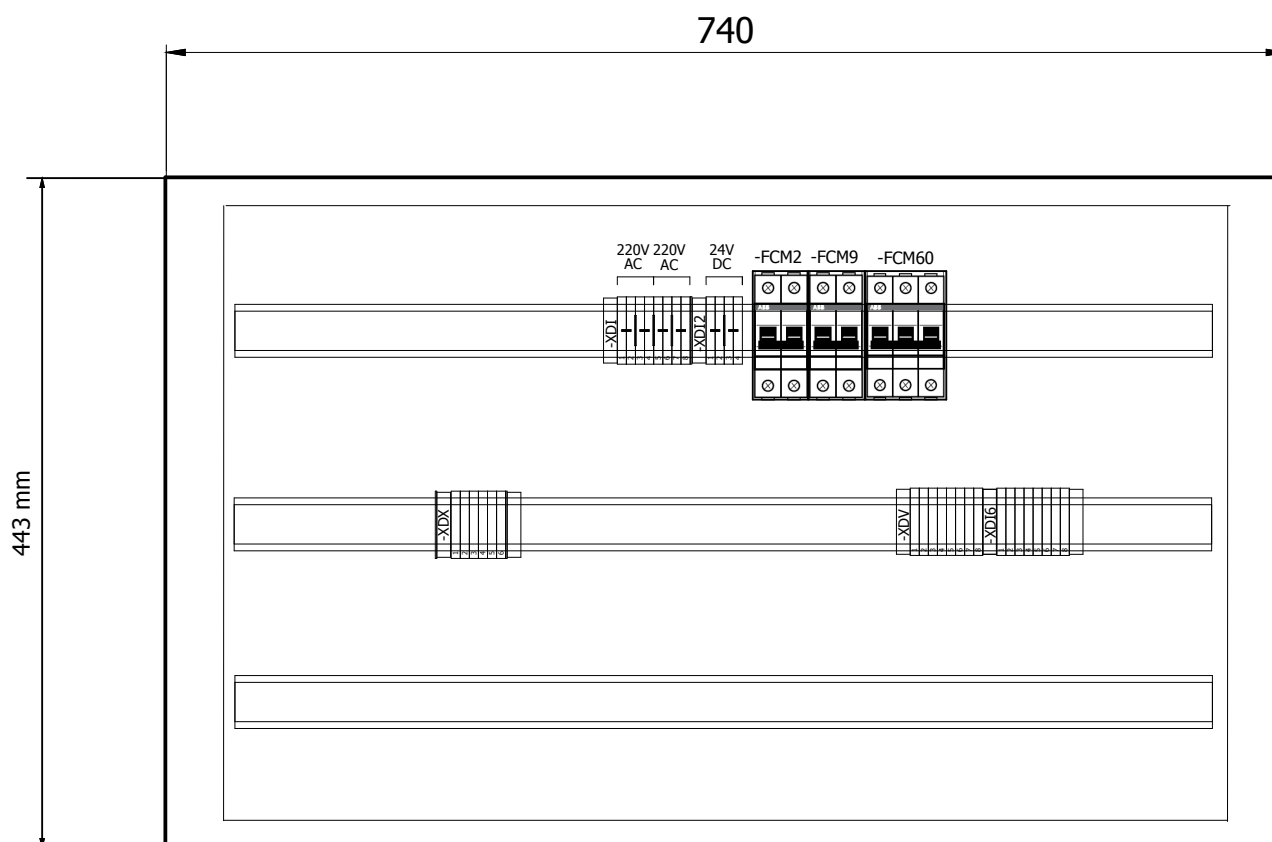
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	MAIN CIRCUITS MVD-20 & MVD-21 20kV Boiler Plant SCHEMATIC DIAGRAM DRC BUS PT	Scale		= H04
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 308
00	Issued for Approval	04.01.19	YR	Checked	AW		Drawing Nbr.	18.042-MNA-308.03		Sh. 8		
Ind.	Revision	Date	Name	Approved	AI	Based on:	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 16
					BE318042							

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[illegible]

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	LV BOX LAYOUT MVD-20 & MVD-21 20kV Boiler Plant SCHEMATIC DIAGRAM DRC BUS PT	Scale		= H04
				Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
00	Issued for Approval	04.01.19	YR	Checked	AW		Drawing Nbr.	18.042-MNA-308.03			Sh.	10
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	Customer Doc. Nbr.				Cont.	16

	1	2	3	4	5	6	7	8
A	Summarized parts list							
	IDABB_F02_005-IDABB							
	Carry over: 0.00							
	TAG ID	DESCRIPTION	MANUFACTUR/BRAND	PART NUMBER	QUANTITY	REMARKS		
	-EB	HEATER 230VAC, 50W	SANYANG	CDXN010033P0009	1	/7.7:E		
B	-FCM2	MCB S202-C2 MCB. 2P, 2A	ABB	2CDS252001R0024	1	/7.2:C		
	-FCM9	MCB S202-C6 MCB. 2P, 6A	ABB	2CDS252001R0064	1	/7.6:C		
	-FCM60	MCB S203-C6 MCB. 3P, 6A	ABB	2CDS253001R0064	1	/8.4:C		
	-PGV	Digital AC Voltmeter Model : DPM PGD3V, 48 X 96 MM	RISHABH	DPM PGD3V	1	/7.2:D		
	-XTV1	Voltage Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting	YONGSUNG	YSPTT-04C	1	/8.5:C		
C								
D								

[illegible]

[illegible]

[illegible]

F		E		D		C		B		A	
REMARK		CABLE	EQUIPMENT	Wire No	Terminal number	TYPE	WIRE NO.	EQUIPMENT	CABLE	REMARK	
==MVD-20_MVD-21=H04+-XDV			-FCM60:1		1	PT 2,5		-BAT1:a		1,5 mm²	
			-FCM60:3		2	PT 2,5		-BAT2:a		1,5 mm²	
			-FCM60:5		3	PT 2,5		-BAT3:a		1,5 mm²	
			-XTV1:8		● 4	PT 2,5		-BAT1:n		1,5 mm²	
					● 5	PT 2,5		-BAT2:n		1,5 mm²	
			-XE2		● 6	PT 2,5		-BAT3:n		1,5 mm²	

[illegible]